

◆ M001 Week 7 — English Worksheet

Topic: Pyramid Variations · **Course:** Pyramid Problems · **Time:** about 45 minutes

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
set size to 7
repeat size times:
  repeat size times:
    if row is 0 OR row is size-1 OR col is 0 OR col is size-1:
      place block down
    move forward
  [turn and walk back to start of next row]
```

Which blocks get placed — all of them, or just some? Where?


```
set size to 9
while size > 0:
  [build one size × size square layer]
  move up by 1
  set size to size - 4
```

How is this pyramid different from a normal one? How many layers?


```
set size to 7
while size > 0:
    [build one HOLLOW size × size layer]
    move up by 1
    set size to size - 2
```

Is this pyramid filled or hollow? Could a person walk inside it?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to build a **hollow layer** — only place a block when it is on the edge of the square (not in the middle).

```
repeat size times:
  repeat size times:
    place block down
    move forward
  [turn and walk back to start of next row]
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent is supposed to build a **stepped pyramid** where each layer is 4 blocks shorter than the one below.

```
set size to 9
while size > 0:
  [build one size × size square layer]
  move up by 1
  set size to size - 1
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent is supposed to build a pyramid with a **gap of 2 blocks** between each layer (the layers float).

```
set size to 7
while size > 0:
    [build one size × size square layer]
    move up by 1
    set size to size - 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Build **one pyramid variation** — hollow, stepped, two-block pattern, or your own idea. Give your variation a **name** and a **one-sentence description**. When you finish, come back here.

Send a photo or video of your variation, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

My variation is called ...

The idea is ...

I changed the ... part of the code to make it different.

The hardest part was ...

To fix it, I ...

If I had more time, I would change ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you walk the camera around your variation. Talk through it like you are teaching someone who has never seen it. Try to use these words: **hollow, stepped, layer, variation, edge.**

1. Walk around your pyramid and show what makes it different.
2. Read your loops out loud and say which part you changed.
3. Show one bug you hit and how you fixed it.
4. Say the name of your variation and what makes it interesting.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.